

JAYASEELA PANDIAN A

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Profile

Electronics and Communication Engineering student specializing in FPGA design, digital verification, and embedded systems. Experienced in Verilog, VHDL, and SystemVerilog with hands-on expertise in designing and verifying digital systems. Strong foundation in RISC-V architecture, MATLAB/Simulink modeling, and real-time hardware implementation. Passionate about developing reliable and high-performance electronic systems.

Projects

Edge AI-Based Smart Disaster and Grid Management System with Custom FPGA Boards

Built an FPGA-integrated Edge AI system for real-time disaster detection and grid management, improving response accuracy and reducing latency in critical environments.

Design and Verification of AMBA APB Protocol using SystemVerilog

Implemented and verified an AMBA APB interface using SystemVerilog with a structured testbench architecture. Built driver, monitor, and scoreboard components to validate read/write operations, protocol timing (Setup & Enable phases), and memory-mapped transactions. Performed functional checks on control and data paths, ensuring correct synchronization of PADDR, PWDATA, PRDATA, and PREADY.

Synchronous FIFO Design and Verification Using SystemVerilog (CRV, Assertions, Coverage)

Designed an 8×8 synchronous FIFO in SystemVerilog and developed a self-checking verification environment using constrained-random stimulus, functional coverage, and assertion-based checks. Validated FIFO behavior across full, empty, overflow, and underflow scenarios using a reference model and automated scoreboard comparison.

Design and Implementation of a Custom Multiply-Accumulate Instruction in an Open-Source RISC-V Processor Core

Designed and implemented a custom MAC instruction in a RISC-V core to enhance computation efficiency, optimizing performance for embedded and AI workloads.

Skills

Hardware Design:

FPGA, Verilog, VHDL, Digital Logic, FSM , UVM, SystemVerilog

Embedded Systems:

RISC-V Architecture, Microcontrollers, IoT Integration

Software Tools & Programming:

Vivado, Arduino IDE, Visual Studio code, EDA playground , Kicad.
C, Python (basics)

Protocols:

AMBA APB, AHB, AXI, UART, I2C

Education

SSLC (10th), Alpha Matriculation Higher Secondary School, State Board (2020)

HSC (12th), Alpha Matriculation Higher Secondary School, State Board (2022)

B.E. – ECE, Vel Tech High Tech Dr. Rangarajan Dr. Sakunthala Engineering College (present)